

Fine-Grained Features in Any-Shot Learning: A Survey

Qiangchang Wang , Wenhao Li , Jinglin Zhang , Zheyun Qin , and Linhao Li 

I. INTRODUCTION

Abstract—Deep learning has achieved remarkable success across a wide range of domains and applications over the past few decades. However, the high performance of most deep learning models is heavily dependent on large volumes of labeled data, which is often labor-intensive, time-consuming, and sometimes impractical to collect. To address this challenge, any-shot learning (ASL)—such as few-shot and zero-shot learning—has been proposed to recognize novel classes with only a few labeled examples or class descriptions. The human visual system excels at identifying discriminative regions in complex scenes, and inspired by this capability, fine-grained features are extracted from input data to emphasize key regions, advancing the field of ASL. Despite the success of these methods, there is a lack of comprehensive surveys on the use of fine-grained features in ASL. This article provides a detailed survey of recent research on fine-grained features in few-shot and zero-shot learning, offering summaries of various approaches, discussing their strengths and limitations, and identifying key challenges, future directions, and research opportunities to drive further advancements in the field. We also explore the potential of ASL in semisupervised learning, weakly supervised learning, and domain adaptation and highlight its applications in computer vision and natural language processing.

Impact Statement—This article surveys fine-grained feature extraction for FSL and ZSL, summarizing recent advances, limitations, and open challenges. We also explore ASL in semisupervised learning, weakly supervised learning, and domain adaptation, and discuss its applications in computer vision and NLP.

Index Terms—Any-shot learning (ASL), few-shot learning (FSL), fine-grained features, local representation learning, part-based modelling, zero-shot learning (ZSL).

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DEEP learning has driven major artificial intelligence (AI) progress by learning complex patterns from large-scale data, but its success still depends heavily on abundant labeled samples. In practice, the long-tailed distribution of categories makes it difficult to collect sufficient data for rare objects, and retraining large models from scratch for new targets is often infeasible in real-time or resource-limited settings.

To alleviate these limitations, any-shot learning (ASL) has emerged, including zero-shot learning (ZSL) and few-shot learning (FSL). ZSL aims to recognize classes unseen during training by transferring knowledge from seen classes to unseen ones [1], [2], [3]. Its core challenge is to identify and integrate transferable knowledge for novel tasks, typically using auxiliary semantic information that encodes class relationships, such as human-defined attributes or text (e.g., class names and descriptions) [2], [3], [4], [5]. In many real scenarios, acquiring a small number of labeled images (e.g., 1–5) for rare categories is possible, motivating FSL [6]. Compared with ZSL, FSL has been more extensively studied [7] and is commonly grouped into transfer learning, generation-based methods, and meta-learning. Transfer learning adapts knowledge from a base dataset to downstream tasks, generation-based methods synthesize additional support examples, and meta-learning learns an initialization that can be rapidly adapted to new tasks via gradient-based updates, typically combining an embedding function, a metric module, and an episodic optimization strategy.

Despite progress, ASL remains challenging. In ZSL, the domain gap between seen and unseen classes often biases predictions toward seen categories, and global feature matching is brittle when seen and unseen images overlap in the global space or when semantic descriptions are highly similar. In FSL, directly transferring image-level global representations from seen to unseen categories can be overly coarse, often missing the fine-grained cues needed for robust generalization under limited supervision.

The human visual system emphasizes important parts of an input while suppressing irrelevant ones, aiding in scene analysis [8], [9], [10]. Inspired by this, researchers have integrated fine-grained features into ASL, including FSL and ZSL. In this survey, fine-grained features refer to localized evidence (regions, parts, patches, and tokens) and the modeling operations built upon them for prediction. We regard a method as fine-grained if it explicitly exploits local units by the following: 1) selecting

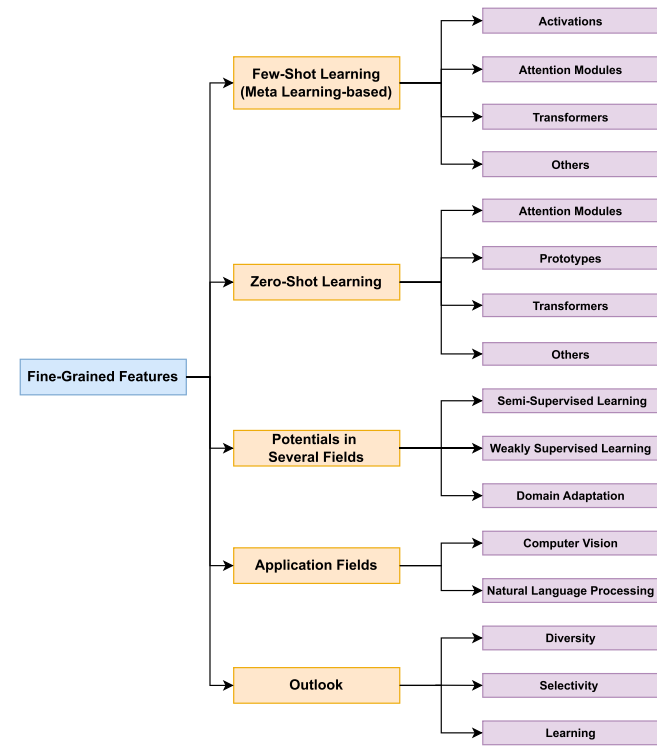


Fig. 1. Taxonomy of fine-grained feature extraction in ASL. Methods are grouped by their dominant fine-grained mechanism on localized units (regions, parts, patches, and tokens), even if multiple components are used. “Others” includes clustering and segmentation and self-supervision. “Potentials” covers extensions beyond ASL, e.g., semi and weakly supervised learning and domain adaptation.

or reweighting local features; 2) establishing region and patch-level correspondences or grounding to mitigate misalignment; and 3) modeling interactions among local units to capture contextual and structural cues. In contrast, approaches dominated by single-vector global pooling and global-to-global matching are treated as coarse-grained baselines unless they introduce explicit local grounding or dense alignment. Fine-grained features offer two advantages: 1) they transfer better than coarse global cues by providing diverse part-level evidence, which is especially useful when novel classes have few labels or limited descriptions; and 2) they are more robust to occlusion, pose, and viewpoint changes than global representations. In ASL, fine-grained methods fall into three main groups: hard part division (fixed crops, stripes, and keypoint-aligned regions), soft part division (attention, segmentation, and clustering to localize informative regions), and Transformer models that align patch tokens for flexible part discovery and long-range interactions.

Despite increasing interest in fine-grained features for ASL, there is no systematic survey categorizing existing methods, challenges, and research directions. This survey provides a comprehensive review of fine-grained features in FSL and ZSL, as shown in Fig. 1, highlighting achievements, current challenges, potentials in specific fields and applications, and promising future directions.

The contributions can be summarized as follows.

- 1) Many important fine-grained features in ASL are summarized. To our best knowledge, this is the first attempt to review the recent progress about different fine-grained features in ASL.
- 2) The summarization about fine-grained features in both ZSL and FSL makes the implementations of fine-grained representation learning easy to understand. The advantages and disadvantages in each field are also discussed.
- 3) We explore the potential of ASL in semisupervised learning, weakly supervised learning, and domain adaptation, with specific applications in computer vision and natural language processing.
- 4) Potential directions of fine-grained features in the future are provided.

The remainder sections of this article is organized as follows. Section II investigates several related surveys. Sections III and IV categorize and review various fine-grained features in FSL and ZSL, respectively. Section V explores the potential of ASL in semisupervised learning, weakly supervised learning, and domain adaptation. Section VI examines some specific application of ASL in computer vision and natural language processing. Section VII discusses future directions of fine-grained features in ASL. Finally, Section VIII presents conclusion.

II. RELATED SURVEYS

Surveys on ZSL [11], [12], [13], FSL [7], [14], and joint ZSL–FSL settings [12], [15] have provided broad overviews from multiple perspectives. For example, [11] groups generalized ZSL into mapping-based, data-augmentation, propagation-based, and optimization-based approaches, and summarizes representative techniques, benchmarks, and applications. Guo et al. [13] review ZSL from an elementwise viewpoint, emphasizing the learning of fine-grained data elements and their relations. On the FSL side, [14] clarifies core concepts (e.g., transfer learning and meta-learning) and organizes common challenges, while [7] categorizes FSL methods from data-, model-, and algorithm-centric views and discusses future directions. In addition, [12] surveys knowledge-graph-aware techniques for both ZSL and FSL, and [15] focuses on ZSL and FSL for node classification in social networks. While these works touch on fine-grained cues, they are not centered on them. In contrast, our survey exclusively targets fine-grained features and provides a dedicated taxonomy of their mechanisms and roles in ASL, enabling a more focused discussion of how local evidence addresses core issues in ZSL and FSL (e.g., domain shift, semantic ambiguity, and low-shot transfer) and how such mechanisms generalize across tasks and domains.

Meanwhile, several surveys focus specifically on attention [16], [17], [18]. Chaudhari et al. [16] review major attention architectures and their impact across applications and interpretability; Guo et al. [17] organize attention in vision into channel, spatial, temporal, and branch variants; and Niu et al. [18] propose a unified view that categorizes attention by softness, input and output forms, and feature representations. Different from these attention-centric reviews, our survey takes a broader fine-grained perspective, covering not only attention but also

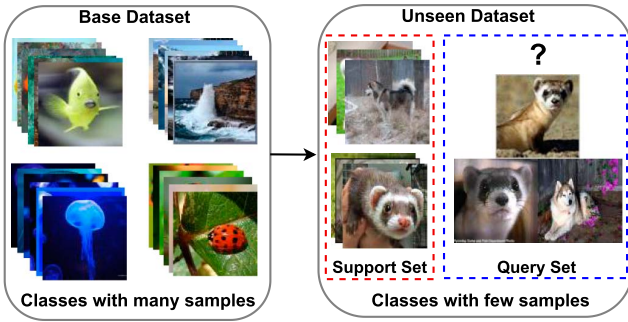


Fig. 2. General pipeline of FSL.

grid and part division, segmentation, clustering, prototypes, and related mechanisms, and analyzes how these fine-grained designs support both FSL and ZSL under a unified ASL lens.

III. FSL

Recent deep learning successes on benchmarks such as ImageNet [19] typically rely on large labeled datasets, which are costly to obtain and hard to scale to new classes. In contrast, humans can recognize novel categories from only a few examples. Motivated by this, FSL aims to classify query samples from unseen classes given only a few labeled examples per class in a support set [6], [20] (Fig. 2). FSL is valuable in data-scarce scenarios (e.g., rare species) and high-cost annotation tasks (e.g., pixel-level labeling). It usually provides 1–5 labeled examples per novel class, far fewer than standard supervised settings. Moreover, FSL evaluates on classes disjoint from training classes, unlike conventional classification where train and test classes coincide.

FSL methods are typically grouped into transfer learning, generation-based, and meta-learning approaches. Transfer learning adapts knowledge from a base dataset, while generation-based methods synthesize extra support examples and are beyond this survey. We focus on meta-learning, which learns an initialization that rapidly adapts to new N -way K -shot tasks. As shown in Fig. 3, meta-training constructs episodic tasks with labeled support sets and query sets for evaluation, and meta-testing adapts the model to episodes from unseen classes.

Many meta-learning methods rely on global pooled features or fully connected classifiers, which may overlook transferable local cues needed for cross-class generalization [21], [22], [23], [24]. Since local patterns are often more consistent across categories, fine-grained modeling is better suited for transfer. Following Section I, we categorize meta-learning-based FSL methods by their dominant operation on localized units (Fig. 1): 1) CNN activations; 2) attention modules for task-conditioned selection; 3) Transformers for long-range patch and token relations; and 4) other mechanisms such as clustering and self-supervision.

A. Activations

Recent studies have harnessed local representations from feature map activations to enhance the performance of FSL.

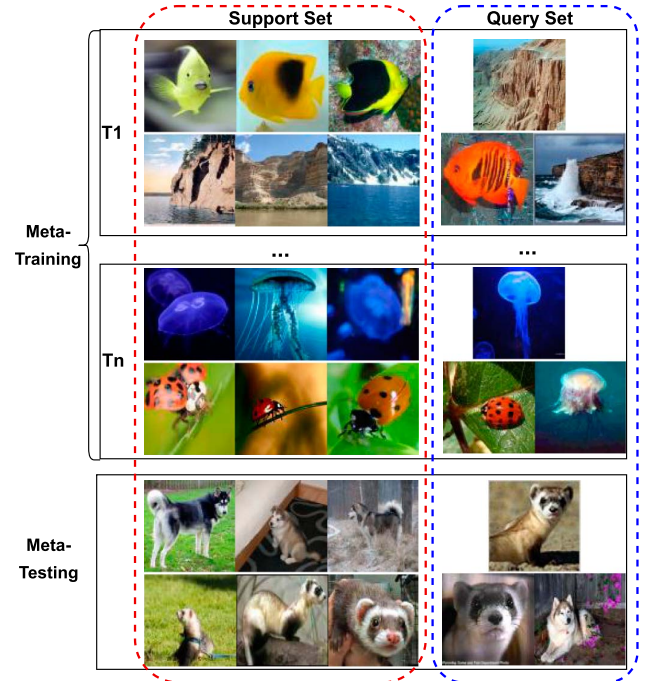


Fig. 3. General pipeline of the meta-based FSL where T_n represents the n th task.

These methods emphasize capturing fine-grained details and spatial information within feature maps, which are essential for distinguishing subtle differences between classes. By leveraging local representations, models can more effectively identify discriminative patterns, even with limited training data, thereby improving their generalization to unseen classes.

Lifchitz et al. [26] leverage local activations for FSL via dense classification over feature-map locations and neural implants for task-specific neurons. As shown in Fig. 4, DN4 [25] matches deep local descriptors with k-NN for image-to-class similarity. DeepEMD [27] uses Earth Mover's Distance for structural matching and a cross-reference module to weight key regions.

Understanding the current task and locating relevant parts of the support set is crucial. A capsule network [28] models spatial relationships and uses triplet loss for discriminative embeddings, with attentive prototypes aggregating support samples based on query reconstruction errors. CNL [29] dynamically captures task-specific features by enhancing features at specific positions and maximizing mutual information. RENet [30] improves meta-learned embeddings by using self- and cross-correlations to capture semantic structures. DeepBDC [31] captures dependencies between local features by modeling joint distributions and measuring discrepancies between joint and marginal functions.

SAML [32] addresses location ambiguity by aligning semantically relevant objects via a “collect-and-select” approach, refining the relation matrix with attention. Xu et al. [33] propose dynamic alignment using a meta-filter for query-region and channel-specific alignment, controlled by a neural ODE. MCL [34] learns bidirectional mutual affiliations, while C2-Net

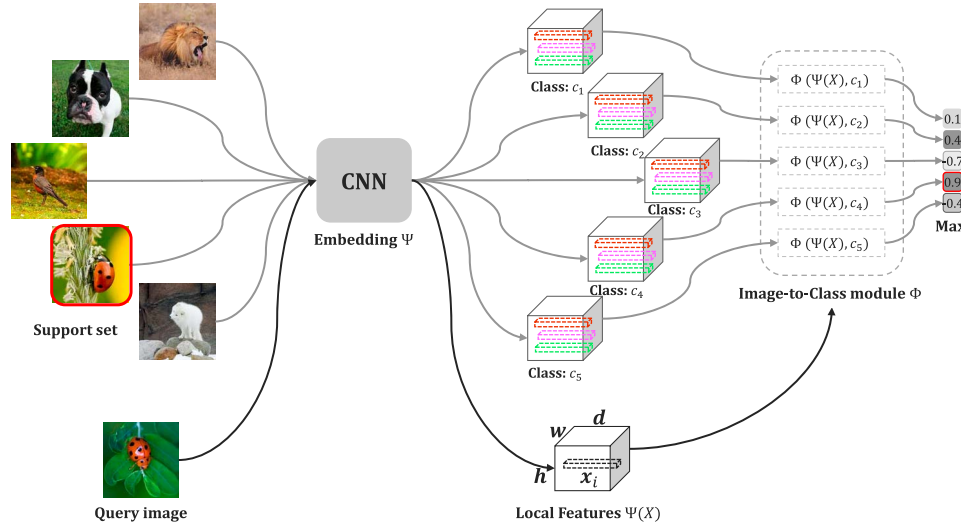


Fig. 4. Quoted from the DN4 [25].

[35] integrates multilayer maps, reduces noise, and recalibrates features with channel activation and positional alignment.

Prior methods characterize appearances at a grid of spatial locations but fail to capture fine spatial details, leading to overfitting to pose or position. To address this, FRN [36] reconstructs a query feature map from support images. MemREIN [37] tackles domain shift using instance normalization to preserve spatial dependencies, with a memorized restitution module and reverse contrastive loss to stabilize feature recovery. SAA [38] generates semantic-aware patch-level soft labels to adaptively upweight class-discriminative regions while suppressing irrelevant background tokens, thereby strengthening fine-grained local discrimination.

DMN4 [39] enhances low-data classification by selecting task-relevant query descriptors and combining mutual nearest neighbor with naive-Bayes nearest neighbor, overcoming limitations of pooled global features and deep descriptors. Inspired by the human ventral stream, [40] introduces hierarchical feature regularization (HFR) and foreground segmentation. HFR models neurons from V1 to V4 to regularize the backbone, capturing hierarchical structures. Foreground segmentation uses masks and edges to distinguish foreground from background, reducing interference. SSP [41] addresses modality gaps in vision-language models by introducing a vision projector for subspace unification and a language projector for class-specific alignment with text features.

Pros and Cons: In FSL, dense activations from pretrained CNN feature maps provide abundant local cues, enabling support–query matching via discriminative parts even for unseen categories. However, these raw features are noisy and background-heavy, and semantic parts are often spatially misaligned across images, which can distort similarity estimation under low-data settings.

B. Attention Modules

Dense activations provide local cues for few-shot matching but are often noisy and misaligned, which can

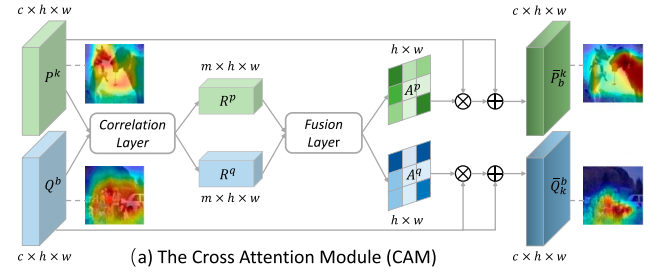


Fig. 5. Quoted from CAN [43].

overwhelm discriminative signals in low-data settings. Attention mechanisms mitigate this by suppressing task-irrelevant regions and establishing soft support–query correspondences, producing more robust, better-aligned representations for accurate few-shot classification.

The FSL problem faces two challenges: 1) semantic objects may vary in position; and 2) fine-grained parts may misalign spatially. PARN [42] uses a deformable extractor and dual correlation attention to filter irrelevant features and align fine-grained ones. CAD [51] leverages self-attention for co-attention, re-weighting spatial attention maps to focus on relevant parts.

Existing methods extract less discriminative features by processing support and query images separately. As shown in Fig. 5, Cross attention network (CAN) [43] uses a cross-attention module to learn semantic relations and localize discriminative parts, with a transductive inference algorithm to generate pseudolabeled query samples. CoAE [44] detects all instances of the same class using co-attention and co-excitation operations to highlight shared discriminative features. SetFeat [53] uses self-attention mechanisms at different convolutional stages for set-to-set matching, predicting the class of a query image from the support set.

Different local patches hold varying importance depending on the task. ATL-Net [45] addresses this with an episodic

attention mechanism and an adaptive threshold-based strategy to select task-aware discriminative patches. MattML [47] uses attention mechanisms in both base and task learners, extracting discriminative parts with convolutional block attention modules. GLoFA [48] tailors task-specific representations using global and local masks for broad and detailed features. TPMN [49] generates task-specific local parts with meta part filters, while TDM [52] enhances class-specific channel information through attention modules, producing task-adaptive feature maps.

Balancing global and local representations is essential for generalization. IPN [46] combines a global stage for support-query relations and a local stage for refinement. To address discriminative location bias, [50] introduces local agnostic training, a similarity method for local feature comparison, and knowledge transfer across classes. Lai et al. [54] propose the clustered-patch element connection layer, clustering similar support patches for semantic consistency and connecting them to query features, with an attention module and distance metric for robust similarity measurement. KTPP [59] adopts a KCA mechanism that keeps only the top-k similar tokens per query to form a sparse attention matrix, while injecting a global context prompt to avoid over-focusing on local details.

Methods often struggle with limited samples and weak feature representation. Chen et al. [55] improve this by using intrablock fusion to combine multilayer features and cross-scale attention to emphasize relevant regions, refining multiscale representations. FeatWalk [56] addresses data scarcity by merging global and local views, generating fine-grained patches. Prototype-guided fusion enhances feature representations by weighting class-relevant views.

Pros and Cons: Attention refines dense activations in FSL by emphasizing task-relevant cues, suppressing background noise, and soft-matching spatial locations to reduce support-query misalignment. However, many attention modules are shallow add-ons to CNN backbones and struggle to capture long-range, multipart dependencies for holistic understanding, motivating more global relation modeling such as Transformers.

C. Transformers

While attention modules improve FSL by refining local features, they are often confined to CNN backbones and thus struggle to capture long-range, holistic dependencies. To address this limitation, researchers turned to Transformers, whose global self-attention directly models relations among all image patches, enabling more coherent representations and stronger few-shot matching.

CrossTransformers [60] mitigates supervision collapse via self-supervised pretraining and a Transformer for part-based comparison and spatial alignment. TransVLAD [66] uses self-supervision to reduce Transformer overfitting and aggregates local cues with NeXtVLAD, while modeling few-shot biases. MG-ViT [64] masks patches to emphasize task-relevant cues and preserves global context, further aided by active sample selection. FewTURE [68] combines masked image modeling with meta fine-tuning and token reweighting to avoid collapse, and

SMKD [70] injects labels into self-distillation with intraclass distillation over class and patch tokens. MAVSI [57] strengthens visual-semantic interaction via spatial and channel attention and contrastive alignment, while MetaFormer [58] improves few-shot adaptation with masked sample attention and patch-grained task attention.

Noisy data in the FSL has a destructive negative effect. TraNFS [62] uses Transformer attention to dynamically reject noisy samples, replacing mean pooling with median or similarity-weighted aggregation. SSFormers [63] splits images into variable-sized patches and applies a sparse spatial Transformer to align query-support patches, followed by patch-level matching for classification.

HCTransformers [61] improve data efficiency with attribute surrogates learning and spectral tokens pooling to address overfitting. SUN [65] accelerates token relation learning in ViTs with location-specific guidance and background path filtration. tSF [67] uses a dataset-attention module to transfer knowledge from the base set to the novel set, filtering semantic features for the target class.

CNN-based methods often produce inaccurate attention maps due to reliance on local features and similar backgrounds. STANet [69] improves this by leveraging global features. SpatialFormer semantic attention highlights semantically similar regions, while target attention focuses on foreground parts of novel classes resembling base classes. As shown in Fig. 6, STANet [69] addresses inaccurate CNN-based attention maps by using global features. The SpatialFormer semantic attention highlights semantically similar regions, while the target attention focuses on foreground parts of novel classes resembling base classes.

Pros and Cons: Transformers benefit FSL by building holistic, context-rich representations via global self-attention, modeling relations among all patches and enabling robust support-query matching beyond isolated local cues. However, they are typically data-hungry and compute-intensive, often relying on large-scale pretraining—an inherent mismatch with the low-data setting of FSL.

D. Others

To reconcile this mismatch, recent work increasingly relies on label-free pretraining. Self-supervised objectives, clustering, and reconstruction on unlabeled data provide scalable supervision, instilling Transformers with general visual priors before episodic adaptation. This alleviates the data bottleneck and yields a strong backbone that can be rapidly adapted from only a few labeled examples.

Part discovery and augmentation network (PDANet) [72] uses a part discovery network and part augmentation to enhance few-shot support examples, improving generalization, as shown in Fig. 7. Wu et al. [74] employs a bi-reconstruction mechanism to increase interclass variations and decrease intraclass variations. ESPT [75] enhances self-supervised learning with random geometric transformations, maximizing consistency in local spatial relationships for better feature transferability.

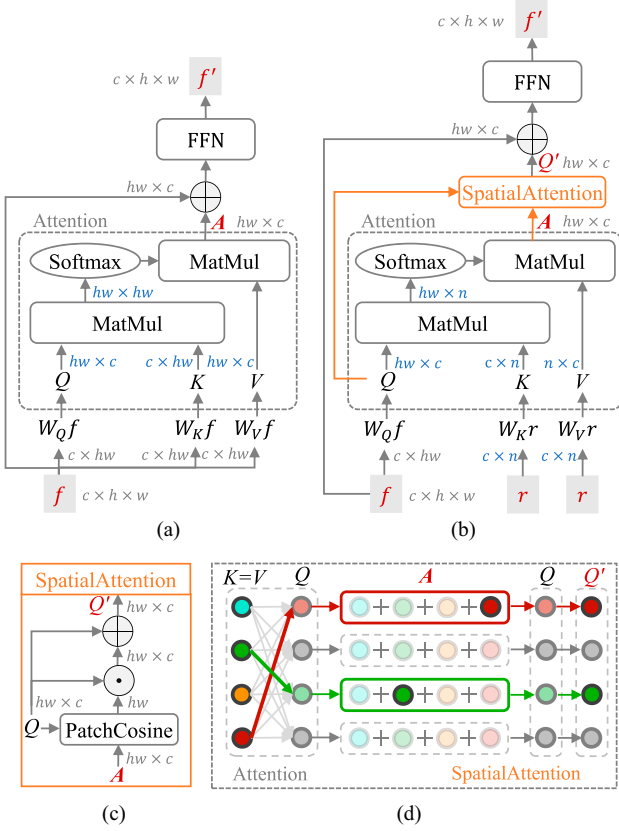


Fig. 6. Quoted from SpatialFormer structure of STANet [69]. (a) Standard transformer. (b) SpatialFormer. (c) SpatialAttention. (d) Intuition for SpatialFormer.

COSOC [73] mitigates background interference by extracting foreground regions, using a clustering-based object seeker with contrastive learning during training and an object concentrator with iterative support-set matching at inference.

CPEA [76] enhances class-relevant representations by iteratively transforming class-agnostic embeddings into class-specific ones via patch interactions, and applies dense patch-level similarity for improved classification.

Pros and Cons: Self-supervised learning, clustering, and reconstruction reduce FSL data scarcity by pretraining backbones on large unlabeled datasets, yielding transferable representations of parts, textures, and structure that can be quickly adapted with only a few labeled support examples.

However, gains hinge on objective–task alignment: reconstruction or clustering may overfit low-level cues or unstable groupings, failing to learn the fine-grained separability needed for downstream FSL. These pipelines can also be costly and sensitive to design choices, creating a pretraining bottleneck that caps few-shot performance.

E. Discussions

The representative FSL models with fine-grained features are summarized in Table I. A good FSL model should be able to overcome the overfitting issue and learn discriminative features from few labeled images. To overcome the overfitting issue, it is necessary to design a simple network which tends to have a

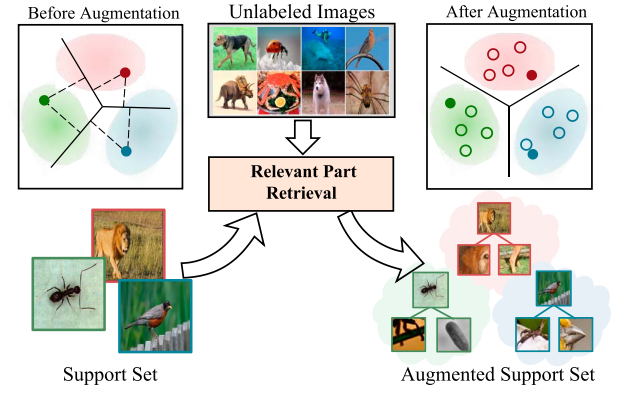


Fig. 7. Quoted from PDANet [72].

weak ability to represent images. Therefore, it is worth exploring how to balance the overfitting issue of models and representational ability of features. Given few labeled images in the FSL, extracting transferable and general local representations can achieve the target.

As shown in Table II, MiniImageNet [78], TieredImageNet [80], CIFAR-FS [81], and CUB-200-2011 [83] are generally used to evaluate the performance of different FSL models. Table III summarizes the result obtained by the aforementioned models.

Besides, although there are possibly only few labeled samples in a novel class, numerous unlabeled images contain rich representations, especially local representations, which can be explored to benefit the FSL task.

IV. ZSL

“What does a zebra look like? It is horse-like with black & white stripes.” Although zebras are not native to China, knowledgeable travelers can describe them by relating them to familiar animals (e.g., horses, tigers, and pandas), helping people who have never seen a zebra imagine and recognize it Fig. 8. Despite the remarkable progress of deep learning in computer vision, most models still rely on large-scale annotated data. In real-world scenarios, however, it is often necessary to recognize novel categories without any labeled images. Motivated by humans’ ability to identify unseen objects from language descriptions, zero-shot learning (ZSL) aims to classify unseen classes by transferring knowledge from seen classes via auxiliary semantic descriptions that bridge the seen–unseen gap, such as visual attributes [84], [85], word embeddings [86], [87], and sentence descriptions [88].

ZSL methods are typically generative or embedding-based. Generative approaches synthesize visual data from unseen-class semantics to reduce seen-class bias, while embedding methods map images into a joint visual–semantic space and classify by nearest-class matching [4], [90]. However, global alignment is brittle under seen-to-unseen domain shift, and whole-image features suffer from: 1) poor separability between seen and unseen samples; and 2) overlapping semantic vectors for similar classes, which hinders fine-grained discrimination.

TABLE I
REPRESENTATIVE FSL MODELS WITH FINE-GRAINED FEATURES

Category	Methods	Venue	Highlights
Activations	[26]	CVPR19	Dense classification on feature maps, neural implants
	DN4 [25]	CVPR19	Deep local descriptors, k-nearest neighbor selection
	SAML [32]	ICCV19	Semantically relevant part alignment
	DeepEMD [27]	CVPR20	Earth mover's distance, cross-reference module
	[28]	ECCV20	Capsule network, triplet loss, attentive prototypes
	CNL [29]	AAAI21	Task-specific and context-aware features, mutual information, task-specific scaling
	[33]	CVPR21	Dynamic meta-filters, neural ordinary differential equation
	FRN [36]	CVPR21	Feature map reconstruction
	RENet [30]	ICCV21	Self-correlational representation, cross-correlational attention
	DMN4 [39]	AAAI22	Mutual nearest neighbor, naive-bayes nearest neighbor
	MCL [34]	CVPR22	Mutual affiliations between query and support features
	DeepBDC [31]	CVPR22	Deep brownian distance covariance
	MemREIN [37]	IJCAI22	Instance normalization, memorized restitution, reverse contrastive loss
	[40]	AAAI23	Ventral stream's neuron, hierarchical feature regularization, foreground segmentation
	C2-Net [35]	AAAI24	Cross-layer feature refinement, cross-sample feature adjustment
Attention Modules	SSP [41]	MM24	Vision projector, language projector
	SAA [38]	ICME25	Semantic-aware patch-level soft labels, position-specific supervision
	PARN [42]	ICCV19	Deformable feature extractor, dual correlation attention
	CAN [43]	NeurIPS19	Cross attention, transductive inference
	CoAE [44]	NeurIPS19	Co-attention, co-excitation, detection
	ATL-Net [45]	IJCAI20	Episodic attentions, adaptive threshold-based selection
	IPN [46]	IJCAI20	Coarse-to-fine cognition, local and global relations
	MattML [47]	IJCAI20	Base and task learners, a gradient-based meta-learning
	GLoFA [48]	AAAI21	Global and local masks, a mask combiner
	TPMN [49]	ICCV21	Meta part filters, adaptive importance generator
	[50]	ICASSP22	Local-agnostic training, local-level similarity and knowledge transfer
	CAD [51]	CVPR22	Cross-relational adaptation
	TDM [52]	CVPR22	Support attention module, query attention module
	SetFeat [53]	CVPR22	Set-to-set matching
	CECNet [54]	IJCAI23	Clustered-patch element connection
Transformers	[55]	AAAI24	Intrablock fusion and cross-scale attention
	FeatWalk [56]	AAAI24	Local view sampling, prototype-guided fusion
	MAVSI [57]	IJCAI24	Spatial cross-attention- and channel attention-based visual-semantic interaction, visual-semantic contrastive learning
	MetaFormer [58]	ICML24	Masked sample attention, patch-grained task attention
	KTPP [59]	MM24	k-NN context attention, coarse-to-fine token filtration
	CrossTransformers [60]	NeurIPS20	Self-supervised learning, part-based comparisons, spatial alignments
	HCTransformers [61]	CVPR22	Attribute surrogates learning, spectral tokens pooling
	TraNFS [62]	CVPR22	Dynamic noise rejection Transformer, median or similarity weighted aggregation
	SSFormers [63]	Arxiv22	Sparse spatial Transformer, image patch matching
	MG-ViT [64]	Arxiv22	Task-relevant mask, residual connections, active learning based sample selection
	SUN [65]	ECCV22	Location-specific supervision, background path filtration, spatial-consistent augmentation
	TransVLAD [66]	ECCV22	Self-supervised learning, NeXtVLAD, supervision bias, simple-characteristic bias
	tSF [67]	ECCV22	Dataset-attention module
	FewTURE [68]	NeurIPS22	Masked image modeling, meta fine-tuning Transformers, inner loop token importance reweighting
	STANet [69]	AAAI23	SpatialFormer semantic attention, SpatialFormer target attention
Others	SMKD [70]	CVPR23	Intraclass knowledge distillation, masked patch tokens reconstruction
	DVLA-RL [71]	ICLR26	Attributes-guided local feature refinement, adaptive self- and cross-attention fusion
	PDANet [72]	IJCAI21	Self-supervised learning, part discovery network, part augmentation network
	COSOC [73]	NeurIPS21	Clustering-based object seeker, shared object concentrator
	[74]	AAAI23	Mutual support-query and query-support reconstruction
	ESPT [75]	AAAI23	Self-supervised learning, episodic spatial pretext task
	CPEA [76]	ICCV23	Self-supervised learning, class-aware adaptation, dense similarity measure
	VT-FSL [77]	NeurIPS25	Iterative refinement to suppress hallucinated attributes, visually grounded fine-grained class descriptions

TABLE II
DETAILS ABOUT DIFFERENT DATASETS USED IN FEW-SHOT IMAGE RECOGNITION

Dataset Name	Year	# Samples	# Categories			Domain
			Train	Valid	Test	
MiniImageNet [78]	2016	60 000	64	16	20	General recognition, derived from ImageNet [79]
TieredImageNet [80]	2018	779 165	351	97	160	General recognition, derived from ImageNet [79]
CIFAR-FS [81]	2019	60 000	64	16	20	General recognition, derived from CIFAR-100 [82]
CUB-200-2011 [83]	2011	11 788	100	50	50	Fine-grained bird recognition

TABLE III
RESULTS (%) OF ONE-SHOT AND FIVE-SHOT TASKS ON MINIIMAGENET, TIEREDIMAGENET, CIFAR-FS, AND CUB-200-2011 USING FSL
MODELS WITH FINE-GRAINED FEATURES

Category	Methods	Venue	MiniImageNet		TieredImageNet		CIFAR-FS		CUB-200-2011	
			One-Shot	Five-Shot	One-Shot	Five-Shot	One-Shot	Five-Shot	One-Shot	Five-Shot
Activations	[26]	CVPR19	62.53	78.95	-	-	-	-	-	-
	DN4 [25]	CVPR19	51.24	71.02	-	-	-	-	53.15	81.90
	SAML [32]	ICCV19	57.69	81.37	-	-	-	-	69.35	81.37
	DeepEMD [27]	CVPR20	65.91	82.41	71.16	86.03	-	-	75.65	88.69
	[28]	ECCV20	66.43	82.13	69.87	86.35	-	-	-	-
	CNL [29]	AAAI21	67.96	83.36	73.42	87.72	-	-	-	-
	[33]	CVPR21	67.76	82.71	71.89	85.96	-	-	-	-
	FRN [36]	CVPR21	66.45	82.83	72.06	86.89	-	-	83.55	92.92
	RENet [30]	ICCV21	67.60	82.58	71.61	85.28	74.51	86.60	79.49	91.11
	DMN4 [39]	AAAI22	66.58	83.52	72.10	85.72	-	-	78.36	92.16
	MCL [34]	CVPR22	69.31	85.11	73.62	86.29	-	-	79.61	90.56
	DeepBDC [31]	CVPR22	67.83	85.45	73.82	89.00	-	-	84.01	94.02
	[40]	AAAI23	69.98	84.22	79.48	91.03	75.78	87.09	82.26	92.60
	C2-Net [35]	AAAI24	-	-	-	-	-	-	78.66	89.43
	SAA [38]	ICME25	78.33	87.09	81.18	90.20	84.64	90.81	-	-
Attention Modules	PARN [42]	ICCV19	55.22	55.22	-	-	-	-	-	-
	CAN [43]	NeurIPS19	63.85	63.85	69.89	84.23	-	-	-	-
	ATL-Net [45]	IJCAI20	54.30	54.30	-	-	-	-	60.91	77.05
	IPN [46]	IJCAI20	67.42	83.98	73.18	86.59	-	-	-	-
	GLoFA [48]	AAAI21	66.12	81.37	69.75	83.58	-	-	-	-
	TPMN [49]	ICCV21	67.64	83.44	72.24	86.55	75.5	87.2	-	-
	[50]	ICASSP22	68.01	68.01	72.27	86.50	78.76	89.60	-	-
	CAD [51]	CVPR22	77.56	77.56	77.55	90.73	79.97	94.13	82.95	90.80
	SetFeat [53]	CVPR22	68.32	82.71	73.63	87.59	-	-	79.60	90.48
	CECNet [54]	IJCAI23	70.20	85.00	73.84	87.36	-	-	-	-
	[55]	AAAI24	88.78	96.76	-	-	78.62	89.15	-	-
	FeatWalk [56]	AAAI24	70.21	87.38	75.25	89.92	-	-	85.67	95.44
	MAVSI [57]	IJCAI24	69.74	69.74	74.61	87.45	80.12	87.13	85.21	91.56
	MetaFormer [58]	ICML24	75.78	90.02	79.05	90.40	81.49	91.91	-	-
	KTPP [59]	MM24	76.71	86.46	80.80	90.01	83.63	90.19	51.59	65.18
Transformers	HCTransformers [61]	CVPR22	74.62	89.19	79.57	91.72	78.88	90.50	-	-
	SSFormers [63]	Arxiv22	67.25	82.75	72.52	86.61	74.50	86.61	-	-
	SUN [65]	ECCV22	67.80	67.80	72.99	86.74	78.37	88.84	-	-
	TransVLAD [66]	ECCV22	68.24	84.42	77.20	84.42	77.48	89.82	82.66	92.45
	tSF [67]	ECCV22	70.23	84.55	74.87	88.05	-	-	-	-
	FewTURE [68]	NeurIPS22	72.40	86.38	76.32	89.96	77.76	88.90	-	-
	STANet [69]	AAAI23	69.86	69.86	74.41	87.64	-	-	-	-
	SMKD [70]	CVPR23	74.28	74.28	78.83	91.02	80.08	90.63	-	-
	DVLA-RL [71]	ICLR26	81.69	88.25	83.02	91.71	87.18	90.59	-	-
	PDANet [72]	IJCAI21	63.84	83.11	69.01	84.20	-	-	-	-
Others	COSOC [73]	NeurIPS21	69.28	85.16	73.57	87.57	-	-	-	-
	[74]	AAAI23	-	-	-	-	-	-	85.44	94.73
	ESPT [75]	AAAI23	68.36	84.11	72.68	87.49	-	-	85.45	94.02
	CPEA [76]	ICCV23	71.97	87.06	76.93	90.12	77.82	88.98	-	-
	VT-FSL [77]	NeurIPS25	83.66	88.38	88.02	91.71	88.67	91.45	57.99	67.68

Recent studies therefore move beyond global representations and exploit fine-grained local features, which transfer better across classes and capture subtle appearance differences under shared attributes. More broadly, fine-grained extraction supports ZSL by providing richer evidence for semantic mapping, enabling part-level knowledge transfer, and mitigating hubness via localized, part-aware regularization. It also complements generative methods by improving the fidelity of synthesized features through part-level supervision. Motivated by these benefits, fine-grained modeling has attracted growing attention in ZSL.

Following the definition in Section I, as illustrated in Fig. 1, existing ZSL methods that exploit fine-grained features can be broadly grouped into the following categories. 1) *Attention Modules*: They can automatically locate discriminative object parts under the supervision of signals, such as attribute information and gaze estimations. 2) *Prototypes*: Several methods learn transferable prototypes (e.g., attribute- and category-level) that generalize from seen to unseen classes. 3) *Transformers*: The

popular Transformers are also explored in this field. 4) *Others*: Other local feature learning methods are also used, including part clustering.

A. Attention Modules

Attention mechanisms can localize discriminative regions without part annotations or detectors. By emphasizing task-relevant areas and suppressing background noise, they improve visual-semantic alignment between seen and unseen classes, enhancing transfer and generalization.

Humans focus on discriminative parts based on class semantics and progressively align them. Inspired by this, stacked semantics-guided attentions (S²GA) [89] assigns progressive weights to regions based on semantics, combining global and local features, as illustrated in Fig. 9. Few works explore the discriminative power of image parts, which outperform attributes in boosting semantic transfer. AREN [91] mines semantic regions with an attentive region embedding stream guided by

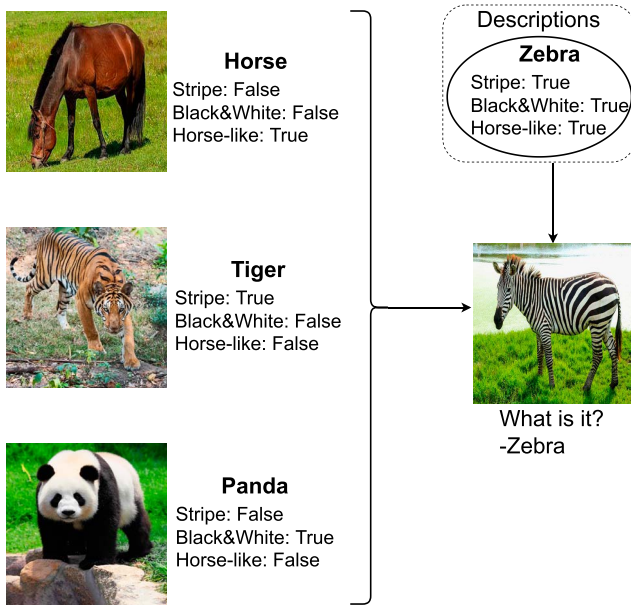


Fig. 8. General pipeline of ZSL.

semantic attributes and an adaptive thresholding module to suppress redundancy. An attentive compressed second-order embedding ensures stable transfer. LFGAA [92] performs object-level attribute attentions for semantic disambiguation. DAZLE [93] learns attribute-based attentions to locate discriminative parts. A self-calibration loss adjusts unseen class probabilities. GEM-ZSL [94] predicts human gaze locations, transforming them to attribute attentions. MSDN [95] uses two subnets for attribute→visual and visual→attribute features, with a semantic distillation loss. CVSC [96] links regional features with attribute vectors, preventing over- or under-prediction of attributes using causal interventions and a conditional correlation loss.

VSE [97] bridges the semantic gap by embedding visual instances into low-dimensional probability matrices and enforcing compact, diverse part attentions. Zhu et al. [98] use semantic-guided multiattention with geometric constraints and losses to learn stronger global and local features. MPGAN [99] generates predefined local patches for ZSL and weights them via voting to reduce background distraction and improve part importance modeling.

RGEN [100] models appearance relations via constrained part attention and graph-based region reasoning [101], with a balance loss to reduce GZSL domain bias. BiAM [102] mitigates feature entanglement in multilabel ZSL using region and scene contextualized blocks. SR2E [103] learns semantic-guided region localization with reinforcement learning and aligns localized regions with semantics via a semantic alignment module.

Pros and Cons: Attention reweights spatial activations to highlight attribute-relevant parts (e.g., zebra stripes or a bird's beak), producing attribute-aware maps that better align visual evidence with unseen-class semantics for improved cross-modal matching.

However, attention is often instance-dependent: maps are generated per image and may focus on different cues across samples, which can yield unstable visual–semantic alignment and weaken class-level consistency, especially for unseen classes with large intraclass variation.

B. Prototypes

Attention can localize attribute-relevant regions, but its image-specific maps may be noisy and inconsistent across samples, weakening class-level alignment for unseen categories. Prototypes mitigate this by building class anchors that aggregate visual or semantic evidence into one representative embedding per class, yielding more stable visual–semantic matching and more consistent zero-shot inference. Prototype-based methods fall into two types: 1) global prototypes that use a single centroid from pooled features; and 2) attribute and part-based prototypes built from localized or attribute-aligned regions. Global centroids are coarse-grained and can obscure subtle cues, whereas attribute and part-based prototypes retain fine-grained semantics via localized grounding, which is our focus.

APN [104] learns local features by regressing attributes, but shared prototypes struggle with image variance, as shown in Fig. 10. DPPN [106] adjusts prototypes dynamically, enlarging category margins for better transferability. Xu et al. [107] improves classification by jointly learning global and local representations, using an attribute prototype network (APN) and a zoom-in module to capture discriminative features.

Pros and Cons: Global prototypes often miss part-level cues and suffer from attribute-binding errors. Fine-grained prototypes instead aggregate part-aligned evidence into stable class anchors, reducing noise and improving visual–semantic consistency.

However, prototypes may still oversimplify a class by collapsing diverse appearances into a single embedding; this “one vector per class” assumption can break for multimodal categories or complex attribute compositions whose distinct visual modes cannot be captured by one prototype.

C. Transformers

Transformers overcome the prototype bottleneck of representing a class with a single vector by using global self-attention to learn context-aware features. Modeling interactions among all patches captures compositional attribute relations and diverse visual modes, enabling more flexible alignment to the semantic space of unseen classes.

As shown in Fig. 11, ZSLViT [105] improves visual–semantic alignment via semantic-embedded tokens and semantic-guided attention, while fusing weakly aligned tokens and filtering irrelevant ones to progressively learn semantic-aware features. TransZSL [116] targets generalized ZSL with adaptive token refinement, hierarchical token aggregation, and semantic-aware prototypes for stronger semantic modeling.

TransZero [108] tackles entangled part features and weak attribute localization with a Transformer-based ZSL design using a feature augmentation encoder and a visual–semantic decoder. TransZero++ [109] strengthens localization and interaction via

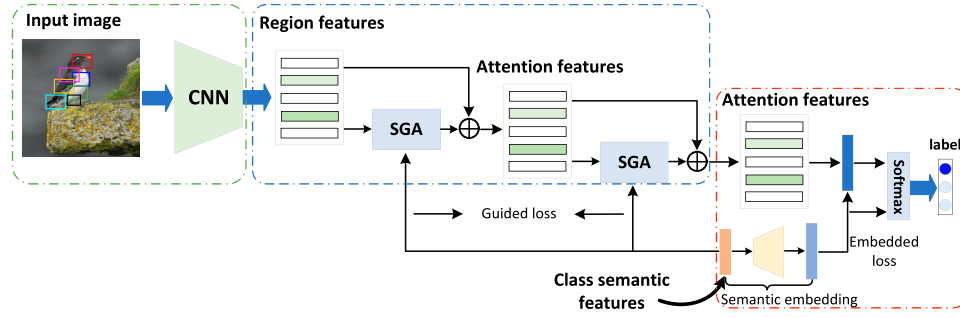
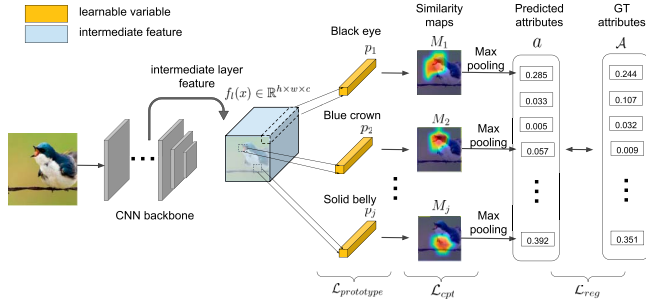
Fig. 9. Quoted from S²GA [89].

Fig. 10. Quoted from APN [104].

an additional visual→attribute Transformer plus feature- and prediction-level semantic collaborative losses. To reduce over-reliance on attributes, DPDN [110] combines attribute-guided part discovery with category-guided mining of complementary discriminative cues.

PSVMA [113] reduces semantic ambiguity by progressively aligning attribute prototypes with visual features, using a Transformer encoder to adapt prototypes and a semantic-guided decoder for cross-domain interaction, plus a debiasing loss. CREST [115] employs visual and attribute grounding Transformers to extract bidirectional regional features for stronger cross-modal interaction. M3R [114] mitigates representation bias via random token mixup and hard cross-modal reconstruction to improve visual–semantic alignment.

Popular ZSL methods rely on human-annotated attributes, which are expensive and difficult to scale. I2Dformer [111] learns image–text alignment from noisy documents by jointly matching global image–document features and local patch-word features in a shared embedding space, using cross-modal attention for fine-grained interactions. DUET [112] injects PLM semantics into ViT and applies attribute-level contrastive learning to mitigate attribute imbalance and co-occurrence.

Pros and Cons: Transformers benefit ZSL by modeling compositional relations among image patches via self-attention, producing holistic, context-aware features that better align with semantic descriptions of unseen classes than single-vector prototype matching.

However, they are typically data-hungry and rely on large-scale pretraining. Moreover, standard Transformers may capture global context without explicitly enforcing semantically meaningful attribute structures (e.g., attribute clusters and cross-image and class relations), which can limit robust transfer to unseen categories.

D. Others

Transformers provide strong but often weakly structured features in ZSL, motivating guided priors. Clustering groups features into semantic patterns, masked image modeling offers label-free pretraining, and graph reasoning encodes interclass and attribute relations. Together, these mechanisms make representations more organized, transferable, and data-efficient for ZSL.

VGSE [117] clusters image regions by visual similarity and links them to unseen classes using word embeddings and a class relation module. Masked unsupervised self-training (MUST) [118] improves CLIP-like models with pseudolabels and self-training, aligning global and local features, as shown in Fig. 12. Chen et al. [119] reduces virtual connections by using explanatory object part aggregation and feature distillation for better generalization.

Pros and Cons: Clustering, masked image modeling, and graph reasoning inject structural priors into ZSL. Clustering groups features into semantic patterns to uncover latent attributes; MIM learns robust representations from unlabeled data; graph models encode and propagate class and instance relations to improve transfer to unseen classes. However, they add training complexity and sensitivity. Clustering is prone to unstable assignments and requires careful tuning. MIM may bias toward reconstruction cues that do not directly serve semantic discrimination. Graph methods rely on graph quality, and noisy edges can propagate errors and hurt generalization.

E. Vision–Language Models (VLMs): From Global Alignment to Fine-Grained Grounding

Large VLMs such as CLIP have reshaped ZSL by shifting the focus from learning mappings to leveraging a prealigned vision–language space. However, standard CLIP inference is largely global-to-global matching between a single image embedding and a text embedding, which is closer to coarse-grained alignment and can miss subtle local cues. In contrast, VLM adaptations such as DenseCLIP [122] and GLIP [123] introduce local dense alignment and token-level cross-modal interactions to explicitly ground language in localized visual evidence. This fine-grained perspective remains important for mitigating attribute binding and spurious correlations and hallucinations by enforcing locality-aware correspondence. Accordingly, in this survey a VLM approach is considered fine-grained only if it

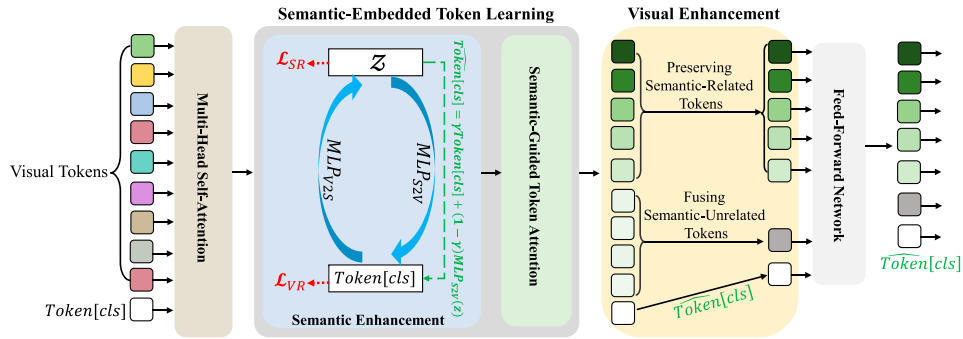


Fig. 11. Quoted from progressive semantic-guided vision transformer (ZSLViT) [105].

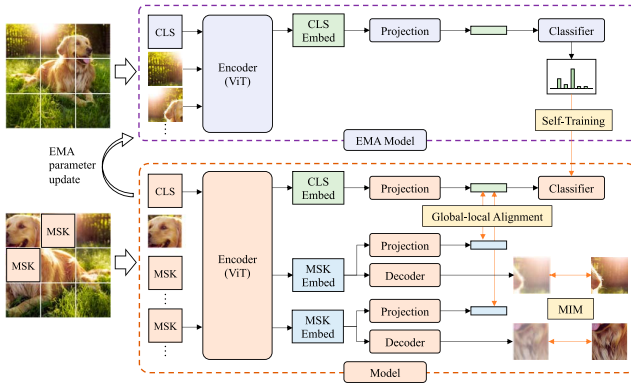


Fig. 12. Quoted from MUST [118].

performs explicit region and patch–token correspondence or dense grounding, rather than relying solely on global embedding matching.

The role of VLMs in ZSL can be summarized as follows. First, as foundational feature extractors, they provide a semantically aligned space learned at web scale [124], [125], reducing dependence on training backbones from limited seen-class data. Second, they enable pure zero-shot inference via direct image–text matching, shifting emphasis to prompt design and calibration for better out-of-distribution generalization [126]. Third, they can be integrated into classical ZSL pipelines by replacing CNN backbones in attribute and prototype frameworks and applying lightweight adaptation (e.g., prompt tuning or selective fine-tuning) to narrow domain gaps with modest compute [127], [128], [129]. Despite these strengths, VLM-based ZSL still faces challenges in specialized domains, inherited biases, and efficiency, motivating future work on more efficient adaptation, principled debiasing, and stronger incorporation of domain knowledge for robust real-world ASL.

F. Discussions

The representative ZSL models with fine-grained features are summarized in Table IV. Most existing works use human-labeled attributes as shared auxiliary information between seen and unseen classes to transfer knowledge from seen classes to unseen categories. Despite their promising performance, the attributes are difficult to annotate and expensive to scale. Unsupervised alternatives worth exploring. Online text documents, such as Wikipedia, can provide rich auxiliary information for ZSL. Besides, large language models which

are trained on large-scale text can also generate high-quality auxiliary class information. However, the granularity between class descriptions and visual cues should be properly aligned.

As shown in Table V, CUB [83], SUN [120], and AWA2 [121] datasets are commonly used to evaluate the performance of various ZSL models. The results achieved by these models are summarized in Table VI.

V. POTENTIALS IN SEVERAL FIELDS

A. Semisupervised Learning

Although ASL (ZSL and FSL) reduces the dependence on large labeled datasets, limited labeled samples and weak auxiliary semantics often provide insufficient supervision, leaving performance far from satisfactory. In settings where only a few labeled samples are available for novel classes but additional unlabeled data can be collected, semisupervised learning offers a natural complement by exploiting unlabeled examples to mitigate label scarcity. Recent ASL + SSL studies mainly improve the reliability of pseudolabels for unlabeled samples to expand the effective training set. Nevertheless, generating accurate pseudolabels remains challenging, especially in extreme low-shot or zero-shot regimes.

B. Weakly Supervised Learning

Most ASL methods still assume access to a small amount of fully labeled data. In dense prediction tasks such as semantic segmentation, conventional supervision requires pixelwise (dense) annotations, which are expensive and time-consuming, particularly in domains such as medical imaging and remote sensing where expert labeling is needed. Sparse annotations provide a cheaper alternative by labeling only a small subset of pixels, but learning under such weak supervision is difficult when labeled data are already scarce, making robust ASL under weak supervision an open challenge.

C. Domain Adaptation

Deep learning models often rely on abundant labeled data, yet many real-world applications have limited or no labels in the deployment domain. Domain adaptation [130] addresses this by transferring knowledge from a labeled source domain to an unlabeled (or sparsely labeled) target domain under distribution shift, typically via distribution alignment. However, many DA

TABLE IV
REPRESENTATIVE ZSL MODELS WITH FINE-GRAINED FEATURES

Category	Methods	Venue	Highlights
Attention Modules	S ² GA [89]	NeurIPS18	Stacked semantics-guided attention
	AREN [91]	CVPR19	Attentive region embedding, adaptive thresholding, attentive compressed second-order embedding
	VSE [97]	CVPR19	A statistical model
	LFGAA [92]	ICCV19	Semantic disambiguation
	[98]	NeurIPS19	A multiattention loss, embedding softmax loss, class-center triplet loss
	DAZLE [93]	CVPR20	Attribute-based features, an attribute embedding, self-calibration loss
	RGEN [100]	ECCV20	Constrained part attention, parts relation reasoning, balance loss
	MPGAN [101]	MM20	Multiple generative models, a voting mechanism
	SR2E [103]	AAAI21	A reinforced region module, a semantic alignment module
	GEM-ZSL [94]	CVPR21	Predict human gaze locations
	BiAM [102]	ICCV21	Region contextualized block, scene contextualized block
	MSDN [95]	CVPR22	Attribute-based visual features, visual-based attribute features, mutual learning
Prototypes	CVSC [96]	MM24	Semantic visual attention module, conditional correlation loss, causal intervention
	APN [104]	NeurIPS20	Attribute prototype, attribute decorrelation
	DPPN [106]	NeurIPS21	Progressive attributes and category prototypes
Transformers	APN [107]	IJCV22	Attribute prototype network, a zoom-in module
	TransZero [108]	AAAI22	A feature augmentation encoder, a visual-semantic decoder
	TransZero++ [109]	TPAMI22	Attribute→visual Transformer, visual → attribute Transformer, semantic collaborative loss
	DPDN [110]	MM22	Attribute-guided part discovery, category-guided part discovery
	I2Dformer [111]	NeurIPS22	Local paired features from noisy documents, fine-grained cross-modal interactions
	DUET [112]	AAAI23	Pretrained language models, cross-modal semantic grounding network, attribute-level contrastive learning
	PSVMA [113]	CVPR23	Instance-motivated semantic Transformer-based encoder, semantic-motivated instance Transformer-based decoder, a debiasing loss
	M3R [114]	MM23	Random token mixup, hard cross-modal reconstruction
	ZSLViT [115]	CVPR24	Progressive semantic-guided vision Transformer, visual enhancement, semantic-embedded token learning
	CREST [115]	MM24	Visual grounding Transformer, attribute grounding Transformer
Others	TransZSL [116]	TCSVT24	Adaptive token refinement, hierarchical token aggregation, semantic-aware prototypes
	VGSE [117]	CVPR22	Image region clusters, class relation discovery
	MUST [118]	ICLR23	Self-training target, masked image modeling, global-local feature alignment
	[119]	TPAMI24	Explanatory graphs, part-net and master-net, feature distillation mechanism

TABLE V
DETAILS ABOUT DIFFERENT DATASETS USED IN GENERALIZED ZERO-SHOT LEARNING

Dataset Name	Year	# Samples	# Categories		# Attributes	Domain
			Seen Set	Unseen Set		
CUB [83]	2011	11 788	150	50	312	Fine-grained bird recognition
SUN [120]	2012	14 340	645	72	102	Fine-grained scene recognition
AWA2 [121]	2014	37 322	40	10	85	Animal image recognition

methods implicitly assume balanced source classes, whereas practical data are frequently long-tailed, with some categories having few or even zero samples. While any-shot supervised domain adaptation has been explored, any-shot unsupervised domain adaptation remains comparatively under-studied.

VI. APPLICATION FIELDS

With growing demands for large-scale data understanding and rapid digital advances, ZSL and FSL have been widely adopted in computer vision and NLP. This section reviews fine-grained features in ASL from a temporal perspective, tracing how FSL and ZSL methods evolve and how they are applied across different tasks and domains.

A. Computer Vision

1) *Image Classification*: Unlike major organizations such as Google and OpenAI, most researchers do not have access to large, high-quality datasets. Image classification is one of the

most prominent applications of ASL methods where it typically involves only a few or even zero samples. We have provided a detailed review of these FSL and ZSL methods in previous sections.

2) *Object Detection*: Object detection aims to localize and classify objects in images or videos, but conventional detectors rely on costly box-level annotations, which are often scarce. Recent zero-shot detection methods have been explored on fine-grained natural images (single- and multilabel) [131], [132], [133], [134], 3D point clouds [135], and specialized domains such as tongue constitution recognition [136], anomaly detection [137], [138], and camouflaged object detection [139]. These methods highlight zero-shot frameworks as a scalable alternative to costly box-level supervision.

3) *Video Understanding*: Recognizing human actions in videos is a key application of ZSL and FSL, motivated by the high cost of annotating complex video data. Fine-grained features help capture subtle motion cues for better action discrimination. DR-CapsGNN [140] models local-global relations

TABLE VI
RESULTS (%) OF ZSL MODELS WITH FINE-GRAINED FEATURES ON THE CUB, AWA2, AND SUN BENCHMARKS

Category	Methods	Venue	CUB			AWA2			SUN		
			U	S	H	U	S	H	U	S	H
Attention Modules	AREN [91]	CVPR19	38.9	78.7	52.1	15.6	92.9	26.7	19.0	38.8	25.5
	VSE [97]	CVPR19	39.5	68.9	50.2	45.6	88.7	60.2	-	-	-
	LFGAA [92]	ICCV19	43.4	79.6	56.2	20.8	34.9	26.1	50.0	90.3	64.4
	[98]	NeurIPS19	36.7	71.3	48.5	-	-	-	37.6	87.1	52.5
	DAZLE [93]	CVPR20	59.6	56.7	58.1	24.3	52.3	33.2	75.7	60.3	67.1
	RGEN [100]	ECCV20	60.0	73.5	66.1	44.0	31.7	36.8	67.1	76.5	71.5
	SR2E [103]	AAAI21	61.6	61.6	65.8	67.5	67.5	39.7	58.0	80.7	67.5
	GEM-ZSL [94]	CVPR21	64.8	77.1	70.4	38.1	35.7	36.9	64.8	77.5	70.6
	MSDN [95]	CVPR22	68.7	67.5	68.1	62.0	74.5	67.7	52.2	34.2	41.3
Prototypes	CVSC [96]	MM24	72.4	78.4	75.3	72.4	78.4	75.3	72.4	78.4	75.3
	APN [104]	NeurIPS20	65.3	69.3	67.2	56.5	78.0	65.5	41.9	34.0	37.6
	DPPN [106]	NeurIPS21	70.2	77.1	73.5	63.1	86.8	73.1	47.9	35.8	41.0
Transformers	APN [107]	IJCV22	67.4	71.6	69.4	61.9	79.4	69.6	40.2	35.2	37.5
	TransZero [108]	AAAI22	69.3	68.3	68.8	61.3	82.3	70.2	52.6	33.4	40.8
	TransZero++ [109]	TPAMI22	67.5	73.6	70.4	64.6	82.7	72.5	48.6	37.8	42.5
	DPDN [110]	MM22	69.1	74.3	71.6	61.9	85.4	71.8	52.4	40.7	45.8
	I2Dformer [111]	NeurIPS22	35.3	57.6	43.8	66.8	76.8	71.5	-	-	-
	DUET [112]	AAAI23	62.9	72.8	67.5	63.7	84.7	72.7	45.7	45.8	45.8
	PSVMA [113]	CVPR23	70.1	77.8	73.8	73.6	77.3	75.4	61.7	45.3	52.3
	M3R [114]	MM23	73.0	77.3	75.1	74.7	87.2	80.4	51.3	44.3	47.5
	ZSLvIT [115]	CVPR24	69.4	78.2	73.6	66.1	84.6	74.2	45.9	48.4	47.3
	CREST [115]	MM24	71.1	72.4	71.7	73.5	87.5	74.1	50.4	39.8	43.2
Others	TransZSL [116]	TCSVT24	72.2	80.1	76.2	69.0	83.4	75.5	46.4	51.8	49.0
	VGSE [117]	CVPR22	21.9	45.5	29.5	51.2	81.8	63.0	24.1	31.8	27.4
Others	[119]	TPAMI24	65.3	66.7	66.0	65.2	79.6	71.7	54.7	59.4	57.0

Note: Performance metrics encompass GZSL accuracies for unseen (U) and seen (S) classes, and the harmonic mean (H) computed as $H = (2 \times S \times U / (S + U))$, which gauges the equilibrium between U and S.

with GNN-based hierarchical capsules, SloshNet [141] uses attention to encode complementary global and local temporal features, and [142] performs hierarchical matching with global, temporal, and spatial alignment to handle coarse-to-fine matching without dense local supervision. These methods successfully bypass video annotation bottlenecks by leveraging multilevel spatiotemporal features.

4) *Image Generation*: From GANs [143] and VAEs to diffusion models, image generation now enables realistic synthesis, and models such as DALL-E [144] and Stable Diffusion [145] can generate images or features for unseen classes in few- and zero-shot settings. Fine-grained cues further improve generation by emphasizing local details (e.g., shape and texture): Xiao et al. [146] use region and boundary-aware cross-attention guidance, StyleGAN [147] provides controllable generation via style vectors, and [148] combines local visual relations with global semantic relations. Therefore, integrating fine-grained local guidance into generative frameworks is crucial for synthesizing highly accurate representations of unseen classes.

5) *Foundation Models*: In recent years, foundation models such as CLIP [124] and SAM [149] have demonstrated strong zero- and few-shot transfer across downstream vision tasks (e.g., classification, detection, and medical segmentation), largely enabled by large-scale pretraining and lightweight adaptation (e.g., fine-tuning). Fine-grained features further boost transfer by injecting localized cues that global embeddings may miss. For instance, SSP [41] selects local features consistent with global semantics to build a vision projector, AnomalyCLIP [138] incorporates local cues via “glocal” context optimization for zero-shot anomaly detection, and H-SAM [150] uses

learnable mask cross-attention with a hierarchical decoder to regulate region interactions and enhance local detail modeling. These approaches demonstrate that augmenting global embeddings with localized features is critical for robust zero-shot adaptation.

B. Natural Language Processing

Recent LLMs have achieved strong NLP performance, from T5 [151] to GPT-3 [152], and can transfer to downstream tasks in a zero-shot manner without task-specific fine-tuning. For efficiency, LongT5 [153] introduces transient global attention over local and global tokens, while COLT5 [154] adopts a two-branch design that routes only salient tokens to a heavyweight branch. Relatedly, GPT-3 employs dense and sparse attention patterns, similar in spirit to Sparse Transformer [155]. By utilizing sparse and localized attention, these models successfully balance strong zero-shot transfer with computational efficiency.

VII. OUTLOOK

In previous sections, fine-grained features have been categorized and summarized. Despite the great success, there is still room to further improve fine-grained feature modeling. Open issues include part redundancy and missing under occlusion, query-support (or visual-semantic) misalignment, and background sensitivity. Inspired by the properties in biological brains, future endeavors may focus on: 1) diversity; 2) selectivity; and 3) learning. For each, we outline actionable goals and evaluation cues to link neuroscience intuition with engineering design.

- 1) *Diversity*: It has been observed in [156] that the primary motor cortex of mice contains over 56 neuronal cell types, and [157] further suggests rich neuronal diversity in the cortex.

Diversity in fine-grained features can improve robustness and data efficiency. In practice, it means learning multiple complementary local representations (parts, patches, tokens, and prototypes) that capture distinct cues with minimal redundancy, helping models handle occlusion, pose and viewpoint shifts, and limited supervision. Beyond implicit diversity [158], [159], [160], [161], future work can enforce it via decorrelation and orthogonality regularizers, coverage constraints, or mixture-of-experts over parts. Progress can be measured with part-usage and overlap and coverage statistics, plus robustness under occlusion and background replacement.

- 2) *Selectivity*: Attention selectivity supports conscious perception by filtering irrelevant inputs and prioritizing salient information. Neuroscience studies suggest that this process involves both spatial control and top-down feedback to select task-relevant neurons and locations [162], [163], [164].

Selectivity in fine-grained vision aims to keep discriminative evidence while suppressing noise and background clutter. In practice, it uses task-conditioned mechanisms to highlight informative regions and tokens and down-weight spurious cues, improving accuracy and efficiency. A key challenge is maintaining stable top-down selection under distribution shifts. We recommend evaluating it with grounding metrics (e.g., deletion and insertion), stability across augmentations, and counterfactual background replacement tests.

- 3) *Learning*: Humans can acquire abstract concepts from very limited examples and generalize them to new contexts [165]. This capability is further supported by cognitive penetrability, where linguistic priors can modulate visual processing to aid recognition under scarce supervision [166].

Future fine-grained modeling should learn transferable local features that generalize across domains and tasks. This can be supported by combining self-supervised objectives that preserve local structure, semantic-guided part discovery and alignment using attributes or text, and uncertainty-aware adaptation to reduce noisy pseudolabels and spurious regions. We recommend evaluating generalization via cross-domain tests, low-shot scaling curves (1→5→10 shots), and robustness to semantic shifts such as seen and unseen confusion or attribute recombination.

VIII. CONCLUSION

Fine-grained features have emerged as a promising research area, enhancing the recognition of novel classes with limited labeled samples or category descriptions while improving robustness to occlusions, pose variations, and viewpoint changes. These features facilitate the effective transfer of knowledge

from seen to unseen classes. This survey has examined various fine-grained techniques in FSL and ZSL, yet numerous challenges and open questions persist. The article also investigates the potential of ASL in critical areas such as semisupervised learning, weakly supervised learning, and domain adaptation, highlighting its applications in both computer vision and natural language processing. By consolidating insights and methodologies from these fields, this review provides a foundation for advancing future research on fine-grained features in ASL.

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